SIMATS ENGINEERING
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ITA0612- MACHINE LEARNING

CO1-AT3 TECHNICAL PRESENTATION

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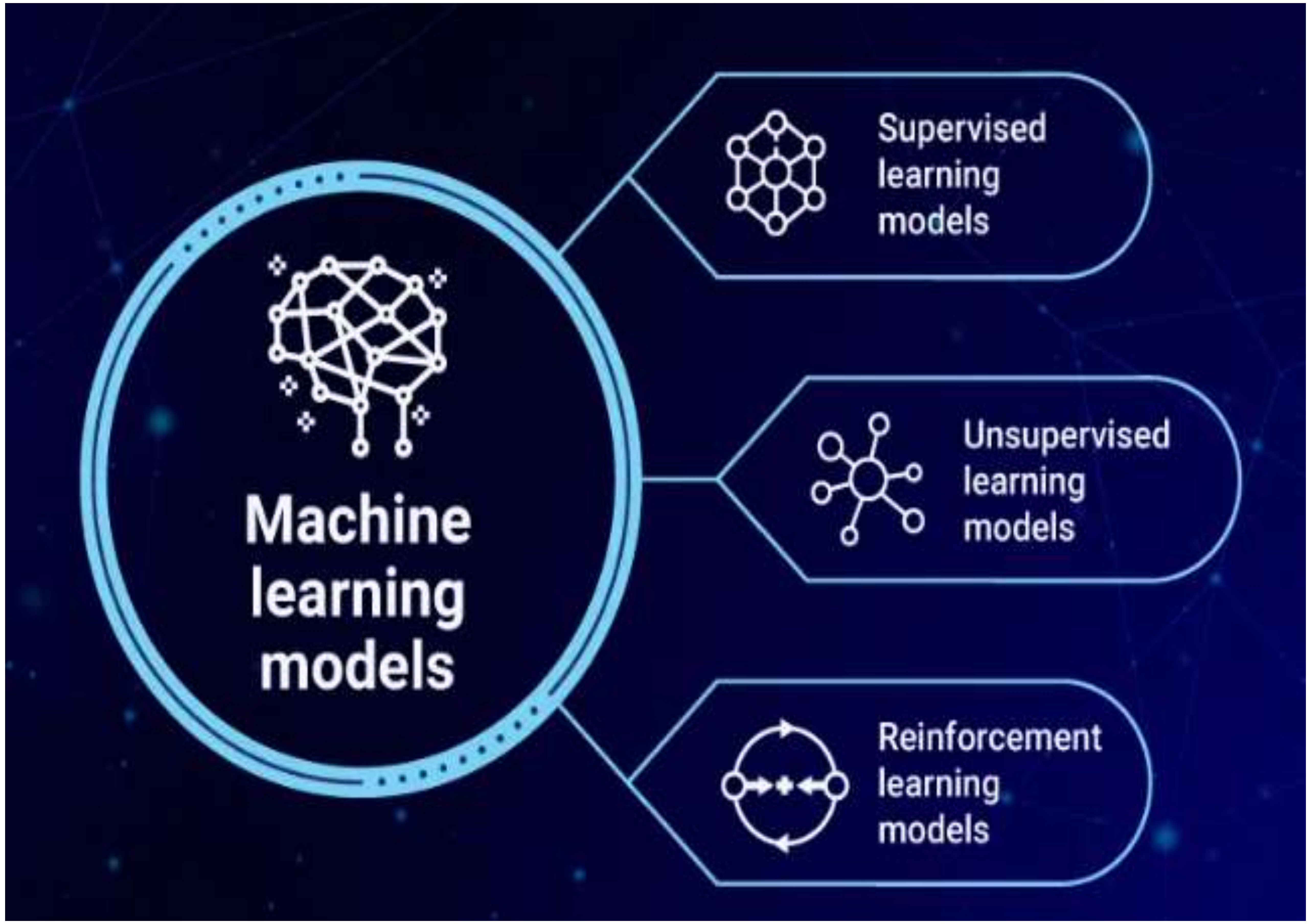
1. Learning Problems and Perspectives

Introduction

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and improve their performance without being explicitly programmed.



Types of Learning Problems

1. Supervised Learning

Uses labeled data (input + correct output).The model learns to predict outputs for new data.

Used for classification and regression.

Examples: Email spam detection, house price prediction.

2. Unsupervised Learning

Uses unlabeled data.Finds hidden patterns or groups in the data.

Used for clustering and association.

Examples: Customer segmentation, market basket analysis.

3. Reinforcement Learning

An agent learns by interacting with an environment.Receives rewards for correct actions and penalties for wrong actions.

Goal: Maximize the total reward.

Examples: Self-driving cars, robotics, game playing.

Perspectives of Machine Learning

1. Symbolic Approach

Represents knowledge using rules and logic.

Easy to understand and explain.

Example: Expert systems.

2. Statistical Approach

Learns from probability and data patterns.

Suitable for large and complex datasets.

Example: Neural Networks, Bayesian Learning.

Key Issues in Machine Learning

- Data Quality
- Poor or missing data reduces model accuracy.
- Clean, relevant data improves performance.
- Overfitting Model learns training data too well.
- Performs poorly on new data.
- Prevented using pruning, regularization, and cross-validation.
- Computational Complexity Large datasets require more processing time and memory.
- Efficient algorithms improve speed and scalability.

CONCLUSION

- The choice of learning method depends on the problem and available data.
- High-quality data, avoiding overfitting, and efficient computation are essential for building accurate and reliable machine learning systems.

2. What is Inductive Bias?

Inductive Bias is the set of assumptions that a machine learning algorithm makes to predict outputs for unseen data.

- Learning algorithms cannot perfectly predict every unseen example.
- They rely on assumptions to generalize from training data.
- Without inductive bias, learning from limited data is impossible.

TYPES OF INDUCTIVE BIAS

1. Restriction Bias

Limits the hypothesis space.

Certain hypotheses are not considered.

Makes learning faster.

Example

Decision Tree algorithm only searches decision tree hypotheses.

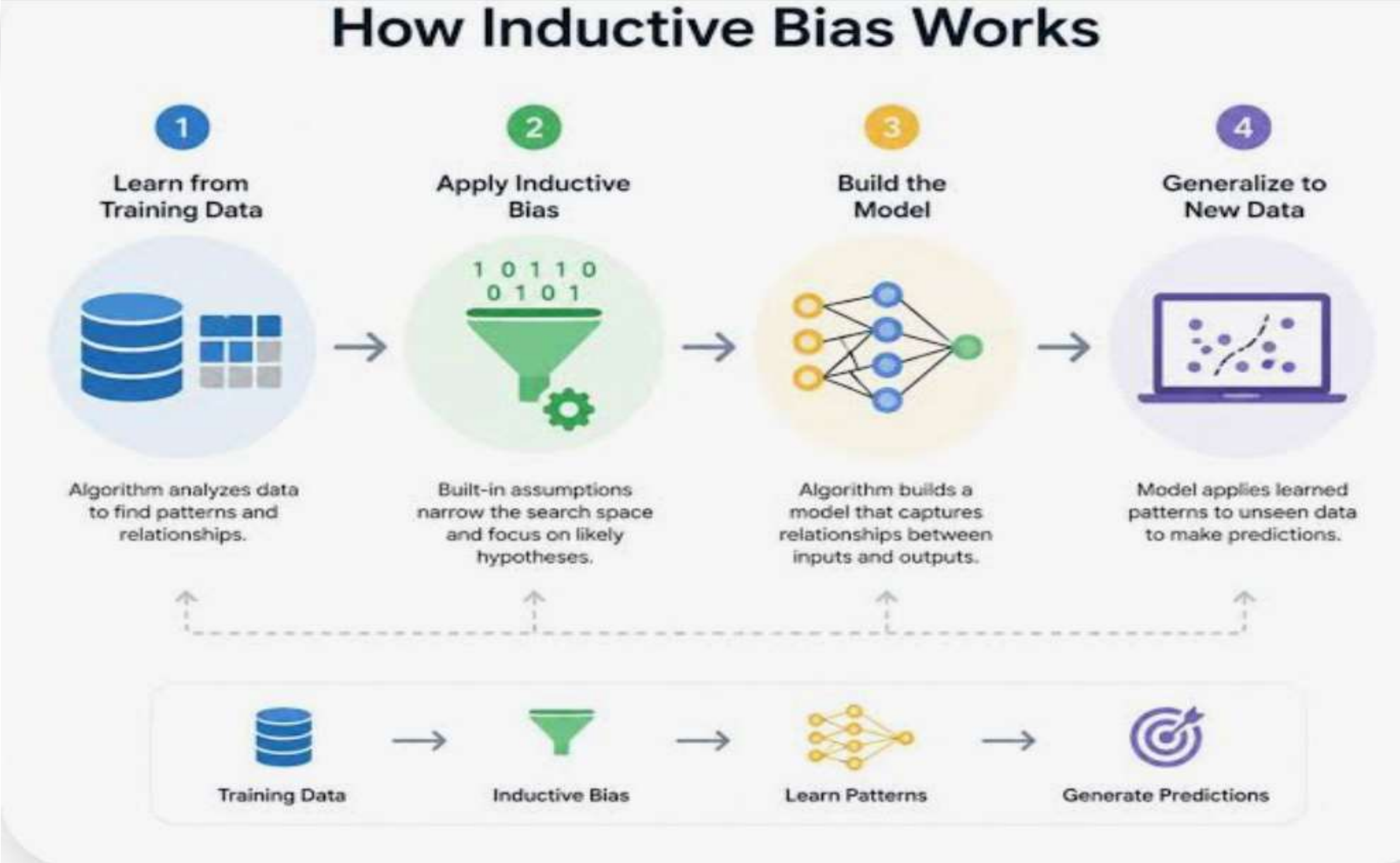
2. Preference Bias

Allows many hypotheses.

Prefers some hypotheses over others.

Example

Choose the simplest hypothesis among many possible solutions.


How Inductive Bias Works
1
Learn from Training Data
2
Apply Inductive Bias
3
Build the Model
4
Generalize to New Data
1 0 1 1 0
0 1 0 1
Algorithm analyzes data to find patterns and relationships.
Built-in assumptions narrow the search space and focus on likely hypotheses.
Algorithm builds a model that captures relationships between inputs and outputs.
Model applies learned patterns to unseen data to make predictions.
Training Data
Inductive Bias
Learn Patterns
Generate Predictions

GENERALIZATION

Generalization means making accurate predictions on unseen data.

Without inductive bias:

Infinite hypotheses fit training data.

Algorithm cannot choose one.

With inductive bias:

Best hypothesis is selected.

Better prediction accuracy.

Improved model performance.

CONCLUSION

- Inductive Bias is essential in machine learning.
- It enables algorithms to learn from limited examples.
- Restriction Bias narrows the hypothesis space.
- Preference Bias ranks hypotheses based on simplicity or other criteria.


THANK
YOU!